

Haysam Elamin

AI Engineer | Machine Learning | Computer Vision | NLP | .NET |

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Professional Summary

Software Developer and AI Engineer specializing in machine learning, computer vision, NLP, and Edge AI systems. Skilled in building end-to-end AI pipelines including data preprocessing, model training, evaluation, optimization, and deployment for real-world industrial and research applications. Experienced working with embedded and Edge AI platforms including Raspberry Pi, NVIDIA Jetson Orin, GPUs, and Hailo NPUs for real-time AI inference and low-latency AI solutions. Passionate about applying AI technologies to solve practical challenges through scalable and efficient intelligent systems. Graduated with High Distinction in Applied Artificial Intelligence and Machine Learning from Conestoga College.

Core Technical Skills

Programming Languages

Python, C#, Java, SQL, JavaScript.

AI & Machine Learning

Machine Learning Pipelines, Deep Learning, Computer Vision, Natural Language Processing (NLP), Model Training, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Edge AI, Real-Time AI Inference

AI Frameworks & Libraries

TensorFlow, Keras, scikit-learn, OpenCV, NumPy, pandas, Matplotlib, CUDA, PyTorch, Hailo Model Zoo.

Edge AI & Embedded Systems

Raspberry Pi, NVIDIA Jetson Orin, GPU Acceleration, Hailo NPU, Embedded AI Systems, AI Model Optimization

.NET Development

.NET Framework, ASP.NET MVC, ASP.NET Core, REST APIs, Entity Framework, Identity Framework, Web API Development, Authentication & Authorization

Web & Software Development

Full-Stack Development, API Integration, Database Design, SDLC Fundamentals.

Databases

SQL Server, MySQL

Tools & Platforms

GitHub, Azure DevOps, Jupyter Notebook, VS Code, Google Colab, Visual Studio

Education

Postgraduate Certificate - Applied Artificial Intelligence & Machine Learning (*High Distinction*) Conestoga College, 2025 – 2026

Professional Experience

Software Developer Jan 2026 - April 2026

Centre for Smart Manufacturing & Advanced Recycling – Conestoga College

- Developed and supported computer vision and AI-based applications for industrial and research use cases within smart manufacturing environments.
- Built and tested machine learning models for object detection, image classification, and real-time inference.
- Worked with Edge AI systems including Raspberry Pi, NVIDIA Jetson Orin, GPU-accelerated platforms, and Hailo NPUs for deploying optimized AI solutions.
- Performed data preprocessing, model training, evaluation, and performance tuning to improve accuracy and efficiency.
- Collaborated with research teams to translate experimental AI models into practical, deployable software solutions.
- Assisted in integrating AI pipelines into real-world workflows for automation and industrial monitoring.

AI Projects

SMARTCV – Smart Computer Vision System (2026)

- Developed a computer vision system designed to detect and classify objects using deep learning models for real-time image analysis.
- Improved object detection performance through dataset preparation and model experimentation using real-world image data.
- Contributed to the development of applied AI solutions for industrial computer vision use cases at the SMART Centre.

AI Medical Patient Simulator – SymptSpher (2025)

Python, DistilBERT, OpenAI API, Streamlit

- Developed an AI medical chatbot that simulates diagnostic conversations for healthcare training environments.
- Implemented transformer-based NLP models to improve contextual response generation.
- Built an interactive web interface enabling real-time AI conversation simulations.

Keyboard Vision AI – AI Child Safety Keyboard (2025)

- Developed a real-time NLP system that detects cyberbullying and harmful language during typing.
- Applied sentiment classification and feature engineering techniques to improve detection accuracy.
- Designed a privacy-preserving architecture that processes text locally on device.

Parkinson's Disease Prediction using Speech Features (2025)

Technologies: scikit-learn, NumPy, Matplotlib

- Trained Naïve Bayes and Logistic Regression models to predict Parkinson's symptoms.